Huawei Access Node Optical Transport Failure Recovery Procedure (ALARM_4517)

Node: IND-ACC-58 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_4517

Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) in the Access Node (IND-ACC-58) at Indiranagar Exchange can occur due to the complex interplay between the optical transmission protocols and the hardware limitations of the Huawei equipment. The SFP/XFP interfaces in the Access Node are responsible for transmitting and receiving optical signals over the fiber link. However, if the SFP/XFP module is degraded or faulty, it can lead to a decrease in the received power, causing the Rx power loss alarm (ALARM_4517). Additionally, fiber cuts or breaks can also result in a loss of signal, triggering the alarm. The Huawei equipment uses the Synchronous Digital Hierarchy (SDH) protocol to manage the optical transport, which involves the use of Virtual Tributary (VT) and Virtual Container (VC) to multiplex and demultiplex the signals. The VT and VC headers contain critical information about the signal characteristics, including the Rx/Tx power levels. If the VT/VC headers are corrupted or mismatched, it can lead to a loss of signal and trigger the ALARM_4517. Therefore, it is essential to perform a thorough diagnosis and resolution of the Optical Transport Failure in the Access Node to ensure reliable and high-quality transmission.

Pre-requisite Checks

- Pre-requisite: Verify the redundant link status and ensure that the redundant link is up and functioning correctly.

- Pre-requisite: Dispatch remote hands to Indiranagar Exchange to physically inspect the fiber optic cables and identify any signs of damage or degradation.

- Pre-requisite: Check the SFP/XFP module status and ensure that it is properly seated and securely connected to the Access Node.

Potential Root Causes

Cause	Probability
Degraded or faulty SFP/XFP module	High
Fiber cuts or breaks	Medium
Corrupted or mismatched VT/VC headers	Low

Diagnostic Commands & Expected Outputs

```
Command: display sfp xfp info, Expected Output: SFP/XFP information:, Description: This command displays the status of the SFP/XFP module, including the module type, status, and Rx/Tx power levels. It helps to identify if the SFP/XFP module is degraded or faulty, which could be causing the ALARM_4517.
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```
Command: display vt vc info, Expected Output: VT/VC information:, Description: This command displays the VT and VC headers, including the signal characteristics, such as Rx/Tx power levels. It helps to identify if the VT/VC headers are corrupted or mismatched, which could be causing the ALARM_4517.

Command: display fiber optic link info, Expected Output: Fiber optic link information:, Description: This command displays the status of the fiber optic link, including the link type, status, and signal characteristics. It helps to identify if there are any issues with the fiber optic link, such as cuts or breaks, which could be causing the ALARM_4517.
```

Resolution Steps

Step 1: Step: Verify the SFP/XFP module status, Command: display sfp xfp info, Description: Verify the status of the SFP/XFP module to ensure that it is properly seated and securely connected to the Access Node. If the SFP/XFP module is degraded or faulty, replace it with a new one.

Step 2: Step: Verify the VT/VC headers, Command: display vt vc info, Description: Verify the VT and VC headers to ensure that they are not corrupted or mismatched. If the VT/VC headers are corrupted or mismatched, re-configure the VT/VC headers to ensure proper signal characteristics.

Step 3: Step: Verify the fiber optic link status, Command: display fiber optic link info, Description: Verify the status of the fiber optic link to ensure that it is functioning correctly. If there are any issues with the fiber optic link, such as cuts or breaks, repair or replace the fiber optic link.

Post-Check Verification

```
Command: verify sfp xfp info, Expected Output: SFP/XFP information: OK, Description: Verify that the SFP/XFP module is functioning correctly and that the Rx/Tx power levels are within the normal range.

Command: verify vt vc info, Expected Output: VT/VC information: OK, Description: Verify that the VT and VC headers are not corrupted or mismatched and that the signal characteristics are within the normal range.

Command: verify fiber optic link info, Expected Output: Fiber optic link information: OK, Description: Verify that the fiber optic link is functioning correctly and that the signal characteristics are within the normal range.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: undo configuration, Description: Restore the original configuration of the Access Node to ensure that the changes made during the resolution process do not affect the normal operation of the Access Node.

- Step: Verify the Access Node status, Command: display access node status, Description: Verify that the Access Node is functioning correctly and that the ALARM_4517 is cleared.